



Sportify

Cloud-Native Multi-Sport Club Management System (MSCMS)

Sportify unifies every operation of a modern multi-sport club — squads, live matches, medical, plan-driven training, real competitions and analytics — into one secure, role-aware platform built on a microservices back-end, a Next.js front-end and a Python ML engine.



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A guided tour through the whole platform

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Problem Definition

Where fragmented club operations break down

Scattered Tooling

Splinters club operations across spreadsheets, WhatsApp threads and paper forms, with no shared system.

Manual Match Ops

Forces staff to log line-ups, events and scores by hand, inviting delays and human error.

Blind Injury Tracking

Records injuries in isolation, with no link to player availability or selection decisions.

No Single Truth

Leaves standings, squads and fixtures duplicated and contradictory across teams and sports.

Problem Definition

More gaps a unified platform has to close



Role-Blind Access

Exposes sensitive data through shared logins, with no per-role permissions or audit trail.



Zero Analytics

Offers no performance insights or match-outcome predictions to guide tactical choices.



Unplanned Training

Schedules sessions ad hoc, with no attendance, drill rating or training-load tracking.



Slow Deal Flow

Drags scouting reports and sponsor negotiations through disconnected emails and back-and-forth.

Project Goals

What Sportify set out to achieve

-  **Unify Club Operations:** Consolidates matches, medical, training, scouting, messaging and finance into one role-aware platform.
-  **Run Matches Live:** Streams each fixture in real time with an accurate, minute-by-minute event timeline and live clock.
-  **Drive Injury Recovery:** Guides every injury from diagnosis through rehab to a final passing fitness test before return-to-play.
-  **Anchor a Single Source of Truth:** Makes the Spring Boot back-end the authoritative store for real fixtures, results and league standings.
-  **Enforce Secure Access:** Applies Keycloak-backed role-based access control across every service, route and action.
-  **Surface Predictive Insights:** Delivers performance analytics and ML-driven predictions to inform line-up, scouting and tactics.



Solutions: Traditional vs. Technological

How Sportify replaces the old club playbook with a cloud-native one



Spreadsheets → Cloud Platform

Replaces scattered, siloed spreadsheets with one unified cloud-native platform that centralizes every workflow.



Paper Files → Digital Journey

Turns loose paper medical records into a staged digital injury lifecycle tied to each player's profile.



No Live Ops → Match Engine

Upgrades silent match days into a broadcast-style live engine streaming events, line-ups and a running clock.



Manual Standings → Real Data

Swaps hand-typed tables for live standings and fixtures ingested automatically from the football-data.org API.



Open Access → RBAC

Ends 'everyone sees everything' by enforcing Keycloak role-based access across players, staff and managers.



Gut Feel → Analytics & ML

Replaces instinct with data dashboards and a Python ML service that predicts outcomes and player insights.



Roles & Access

How each role sees a tailored, secure surface



Administrator

Key Access: every dashboard route, including admin-only User Management.

- Primary Functions: manages all accounts and roles and oversees the platform with unrestricted access.



Sport Manager

Key Access: Club Hub, Teams, Staff, Players, Matches, Scouting, Contracts and Sponsors. • Primary Functions: runs a sport end-to-end, from squad and staff to transfers and sponsor deals.



Team Manager

Key Access: Club Hub, Teams, Staff, Players, Training, Matches, Contracts, Reports. • Primary Functions: coordinates a team's roster, training schedule, fixtures and contract logistics.



Head Coach

Key Access: Players, Teams, Training, Matches, Medical, Analytics, Reports, Alerts. • Primary Functions: leads coaching, sets up training and line-ups, and reviews performance and fitness.



Roles & Access

From analysts and scouts to read-only players & fans



Doctor / Physiotherapist

Key Access: Medical, Players and Training (doctor also gets Alerts). • Primary Functions: manages medical profiles and the injury lifecycle, monitoring availability for training.



Performance Analyst

Key Access: Analytics, Training Analytics, Reports, Matches, Players and the ML Predictor & Rating tools. • Primary Functions: analyzes match and training data and runs AI predictions and ratings.



Scout

Key Access: Scouting & Scouting Ops, Players, Matches, Competitions, Head-to-Head, Reports. • Primary Functions: tracks and evaluates prospects, logging scouting reports against players and fixtures.



Player / Fan

Key Access: read-only Club Hub, Players, Matches, Competitions and Head-to-Head (players also see Training and Medical). • Primary Functions: browses club info and follows matches in a view-only UI.



System Architecture

A cloud-native Spring Cloud microservice mesh behind one JWT-secured gateway



Next.js Front-End

Serves the React dashboard and talks to the platform through a single API Gateway origin.



Spring Cloud Gateway

Verifies Keycloak JWTs and load-balances every request to the right service on port 8080.



Six Domain Services

Splits into user (8081), player (8084), training-match (8087), medical (8086), reports (8083) and notification (8085), each with a private database.



Eureka + Config

Registers services in Eureka (8761) and feeds them centralized settings from the Config Server (8082).



Keycloak Identity

Issues and signs OAuth2 tokens for the mscms realm that every service validates as a resource server.



Kafka Event Backbone

Streams domain events asynchronously via an outbox publisher so notifications and analytics never block.



FastAPI ML Engine

Exposes Python match-prediction and player-rating models that the gateway routes to under /ml.



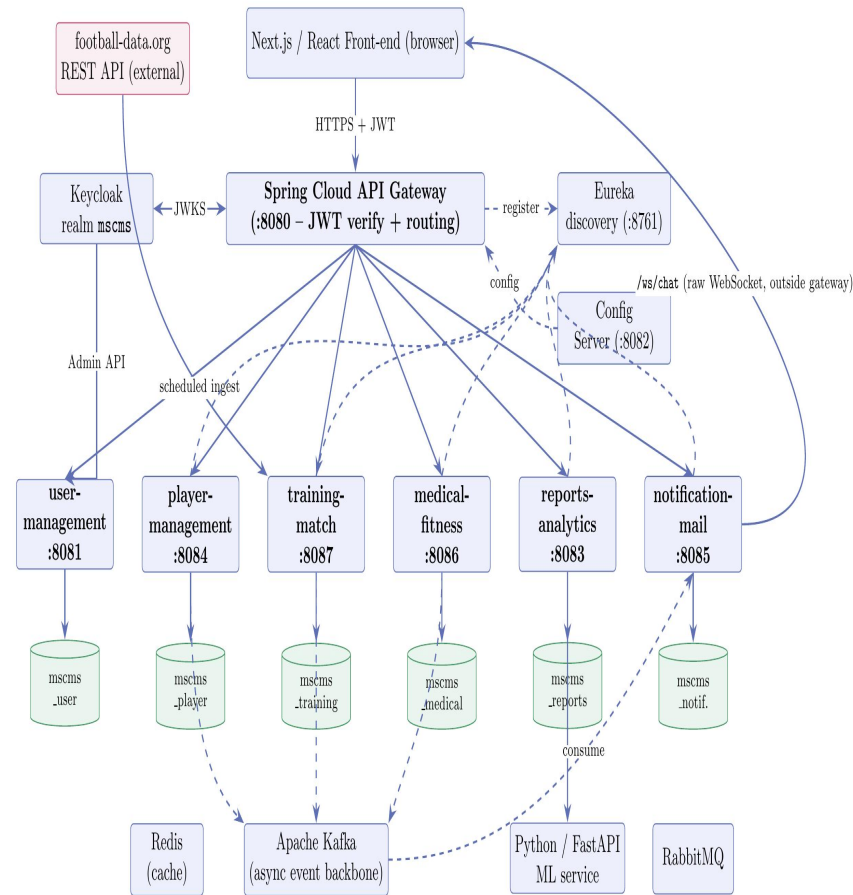
football-data.org Feed

Pulls real La Liga and Champions League fixtures and standings on a scheduled sync into the match service.



Architecture at a Glance

One gateway, six domain services, an event backbone and an ML engine



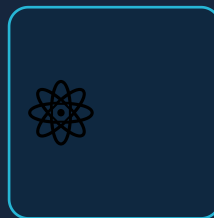
Enterprise-Grade Tech Stack

Proven, production-grade technology at every layer



Back-End

Java 21 / Spring Boot 3.5
Spring Cloud Gateway + Eureka
PostgreSQL 14 (DB per service)



Front-End

Next.js 16 (App Router)
React 19 & Tailwind v4
WebSockets for real-time



Infrastructure

Keycloak 26 (Identity / JWT)
Apache Kafka (Eventing)
Docker · Python/FastAPI ML



Key Features

Each capability, one at a time — with a real screen of it in action.



Role-Aware Dashboards

One adaptive surface per role — from administrator to fan.



Adaptive surface — Every menu, page and action is tailored to the signed-in role.



Enforced twice — A shared permissions map gates the sidebar and the server-side route guard.



Fourteen roles — From administrator to fan, each account sees only what it should.



Role-Aware Dashboards

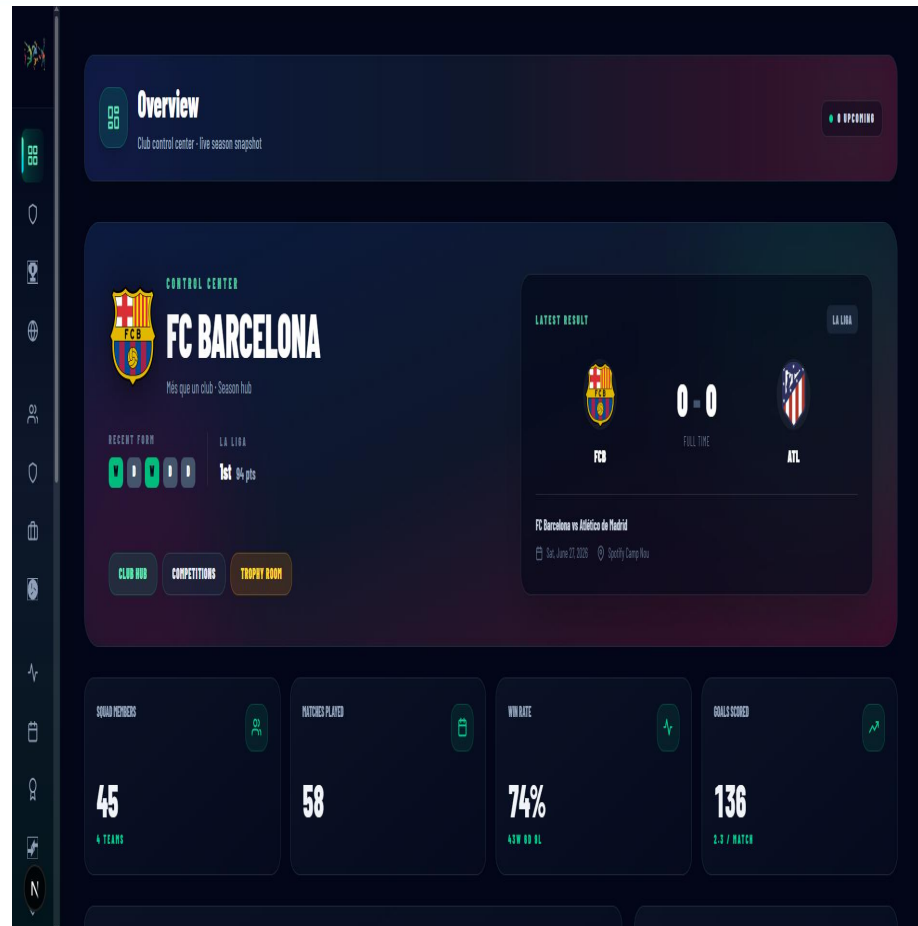


fig-dashboard.png



Multi-Sport & Multi-Team

Sports, teams, rosters, contracts and transfers in one place.



Many sports — Football, basketball, handball, swimming and more, all in one system.



First & reserve squads — Each sport runs a senior team plus a reserve / academy team.



Full squad data — Players, staff, contracts and transfers managed per team.



Multi-Sport & Multi-Team

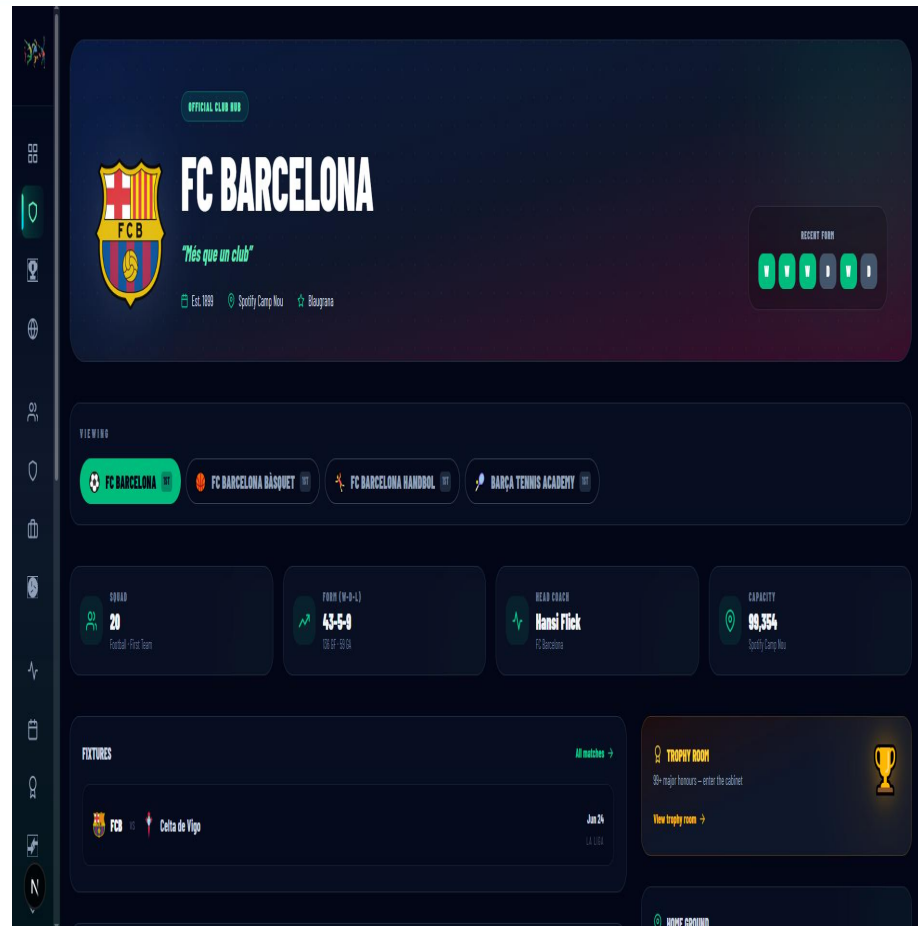


fig-club-hub.png



Real Competitions

Live La Liga and the new-format UEFA Champions League.

E S

La Liga Live — Shows live La Liga standings and fixtures, with results that update automatically as matches are played.



New-Format Champions League — Renders the redesigned UCL as one 36-team league-phase table with qualification zones, then unlocks the knockout bracket.



World Cup 2026 — Presents the 48-team World Cup across 12 group tables and a knockout bracket, seeding the qualified teams into the Round of 32 when the draw is TBD.



Head-to-Head — Builds FC Barcelona's all-time win-draw-loss record, goals and win-rate against every opponent from finished matches.



Real Competitions

← ALL COMPETITIONS

LEAVE 2024/25 FOOTBALL COMPLETED READ-ONLY DELETE

La Liga
20 YEARS - 300/300 MATCHES PLAYED

CHAMPION
FC Barcelona

MATCHES STANDINGS

STANDINGS

#	TEAM	P	W	D	L	GF	GA	GD	PTS
1	FC Barcelona	38	31	1	6	95	36	+59	94
2	Real Madrid CF	38	27	5	6	77	35	+42	86
3	Villarreal CF	38	22	6	10	72	48	+26	72
4	Club Atlético de Madrid	38	21	6	11	62	44	+18	69
5	Real Betis Balompí	38	15	15	8	59	48	+11	60

fig-laliga.png



Live-Match Engine

A broadcast-style match-day cockpit, run minute by minute.



Phase State Machine — Walks every match through scheduled → first half → half time → second half → full time, auto-going-live at kickoff.



Phase-Bound Events — Records goals, cards, subs and injuries against a live clock, clamping each event's minute to the phase in play.



Knockout Decider — Triggers two extra-time periods and an alternating penalty shoot-out only when a true knockout tie is still level.



Live Persistence — Streams every score change, event and status flip to the training-match service so the timeline survives a reload.



Live-Match Engine

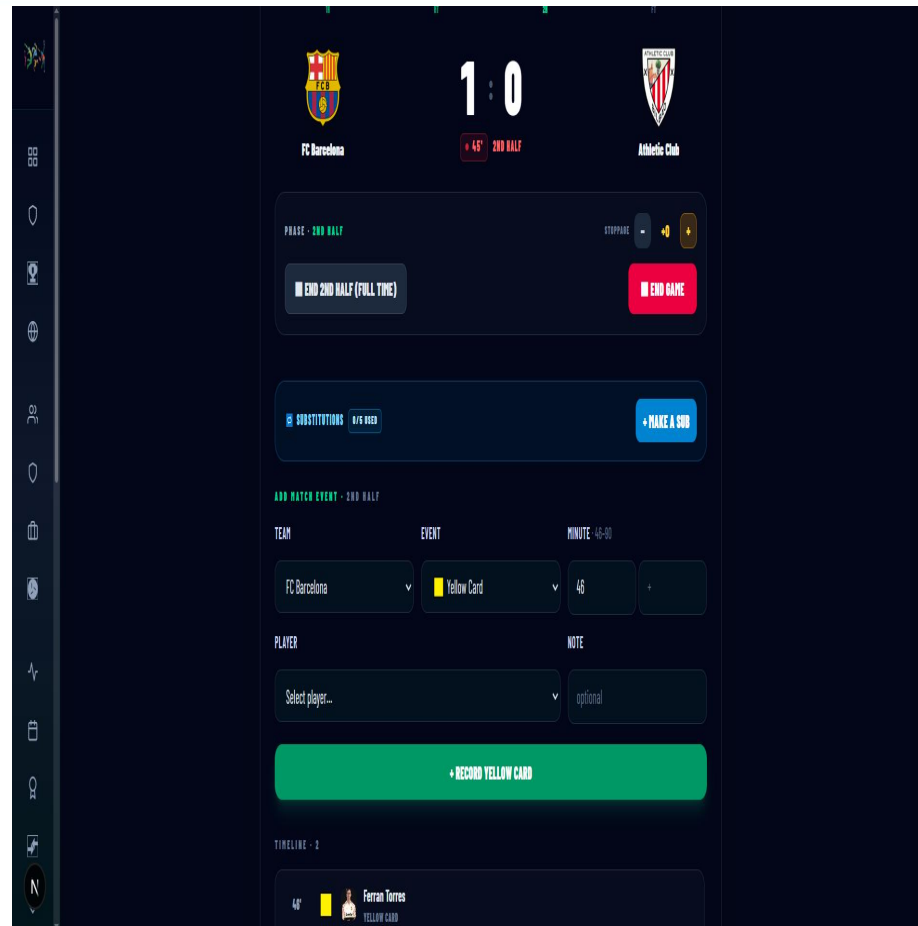


fig-live-match.png



Advanced Injury Management

A medically-governed return-to-play pipeline.



Staged Injury Lifecycle — Drives every injury through REPORTED → DIAGNOSED → TREATING → RECOVERING → RECOVERED as the status that gates each next action.



Single Source of Truth — Makes the medical service the one authority on availability, so card, stepper and squad all read the same status.



Auto-Removal When Injured — Pulls an injured player out of match line-ups, training and attendance the moment the injury is logged, and notifies the team.



Mandatory Fitness Test — Requires a passing return-to-play fitness test before the player flips to RECOVERED and is restored as AVAILABLE.



new-medical-lifecycle.PNG



Plan-Driven Training

From season plan to scored session, one squad at a time.



Training Plans — Anchors one plan to a single team and sport, with a start-to-end date range, training type, goals and focus that every session inherits.



Scheduled Sessions — Places each session at its own date and time inside the plan's range, then lists them in chronological order with a done count.



Ordered Drills — Stamps each drill with a category, duration and sequence, and rejects the session if the drills' total minutes reach the session length.



Squad Attendance — Records a per-player status and offers only non-injured players who belong to that plan's team for call-up.

Plan-Driven Training

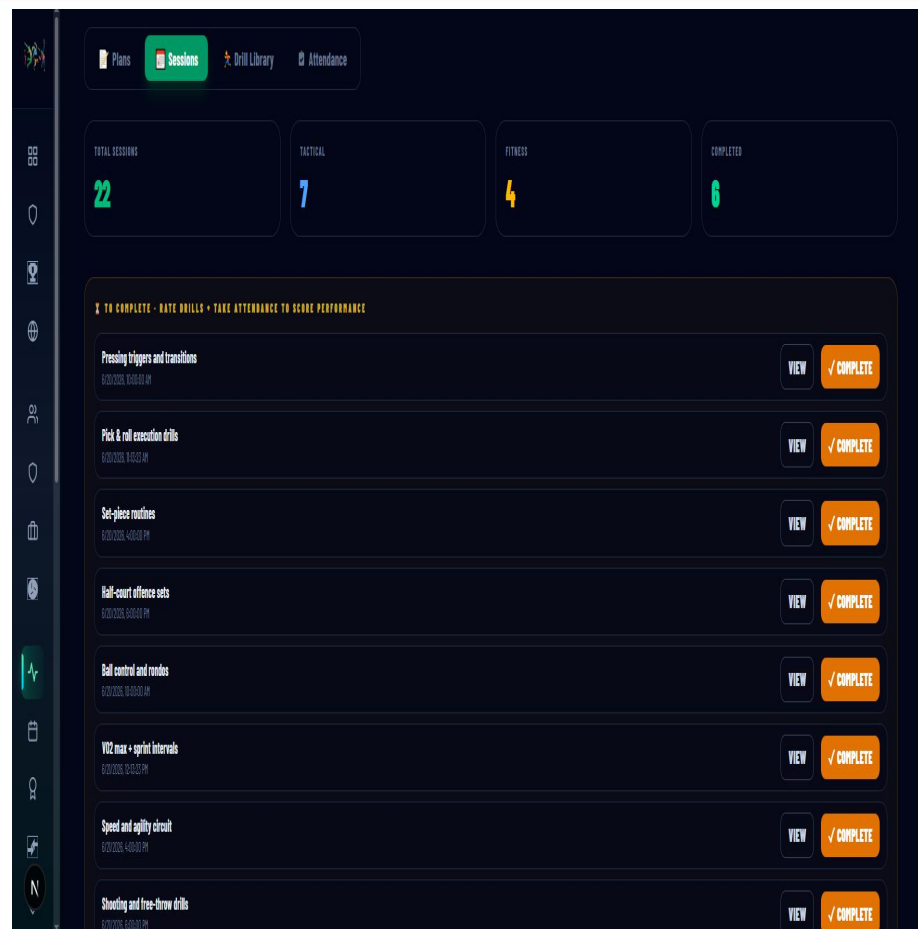


fig-training.png



Scouting & Sponsorship

Scout reports on prospects and sponsor-offer negotiation.



Sign-or-pass reports — Scouts log structured reports on external prospects.



Sponsor negotiation — Offers move through pending, stale and accepted states.



Tied to fixtures — Reports attach to the players and matches they were scouted at.



Scouting & Sponsorship

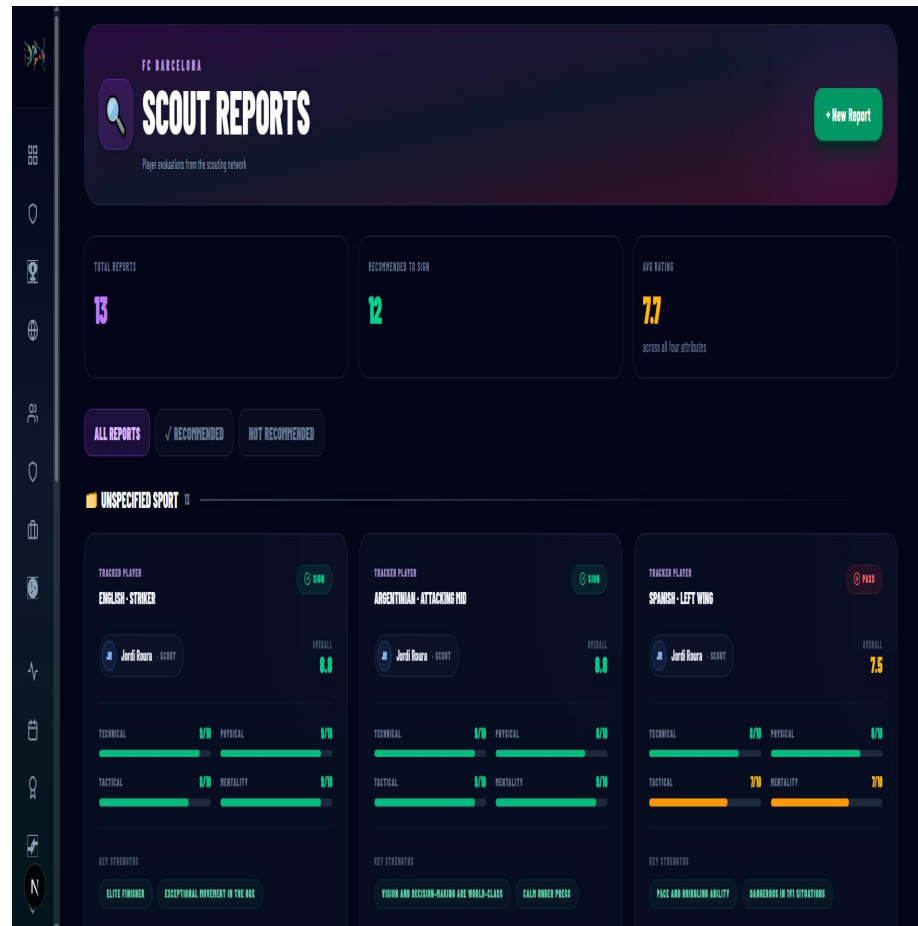


fig-scouting.png



Real-Time Messaging

Club-wide group chat over a live socket.



Live WebSocket — Messages, reactions, replies and read receipts in real time.



Polling fallback — Falls back to polling when the socket is unavailable.



Team-scoped — Staff and players only — fans stay on the spectator surface.

Real-Time Messaging

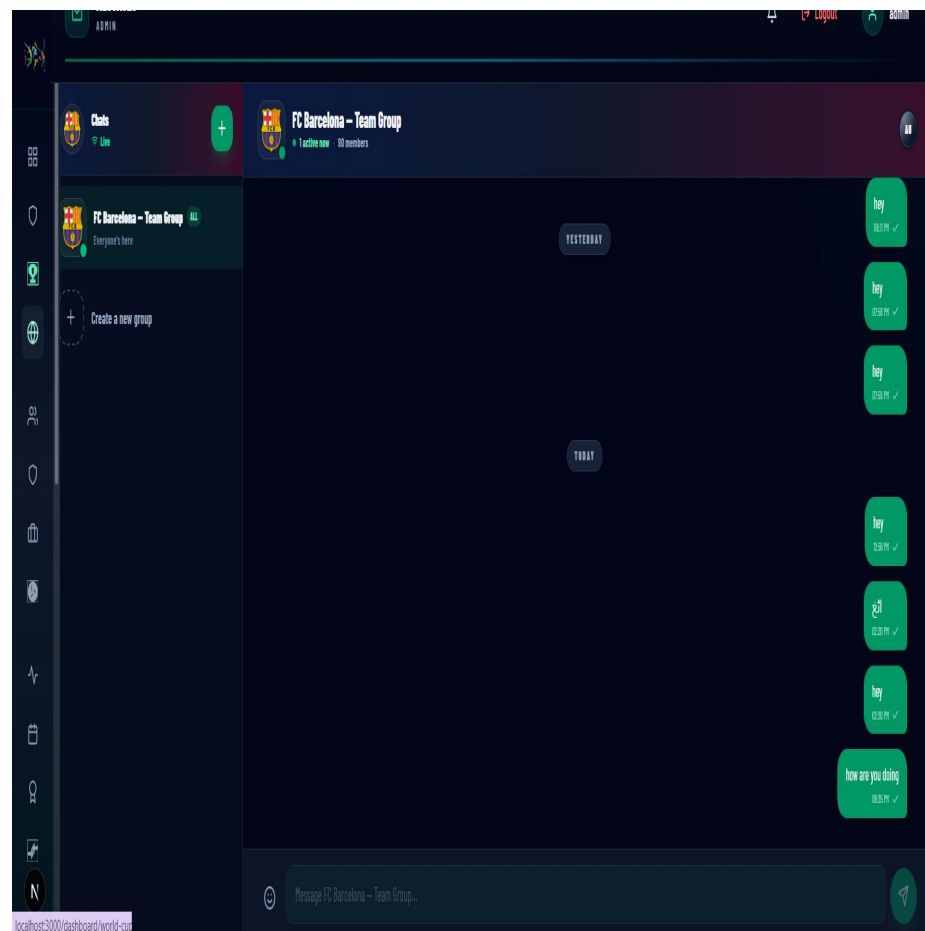


fig-messages.png



Analytics & ML Predictions

Match-outcome and player-rating predictions on real data.



KPI dashboards — Aggregates team, player and training KPIs.



Match predictor — A Python/FastAPI service forecasts match outcomes.



Player ratings — ML 0–100 ratings inform selection and scouting.



Analytics & ML Predictions

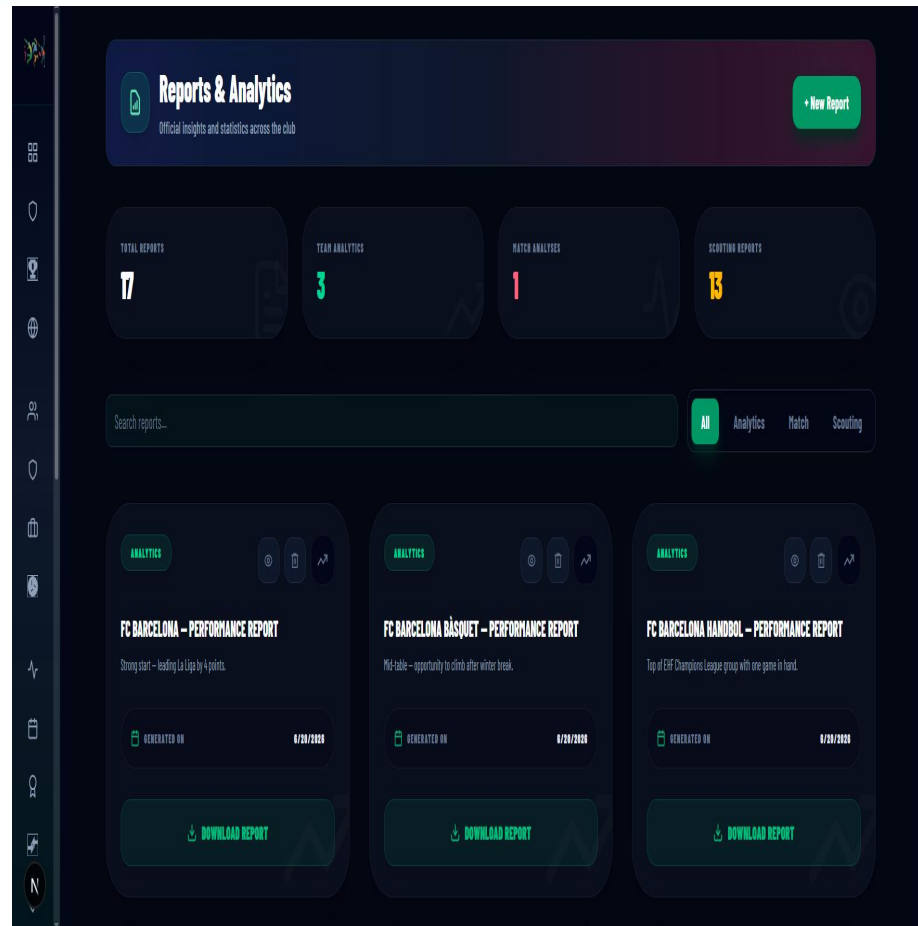


fig-reports.png



Live Notifications

The whole club, kept in the loop in real time.



Top-bar bell — Injuries, goals, transfers and report-ready events surface instantly.



Event-driven — Kafka events fan out to the notification service.



Alerts center — A full history of everything that happened.

Live Notifications

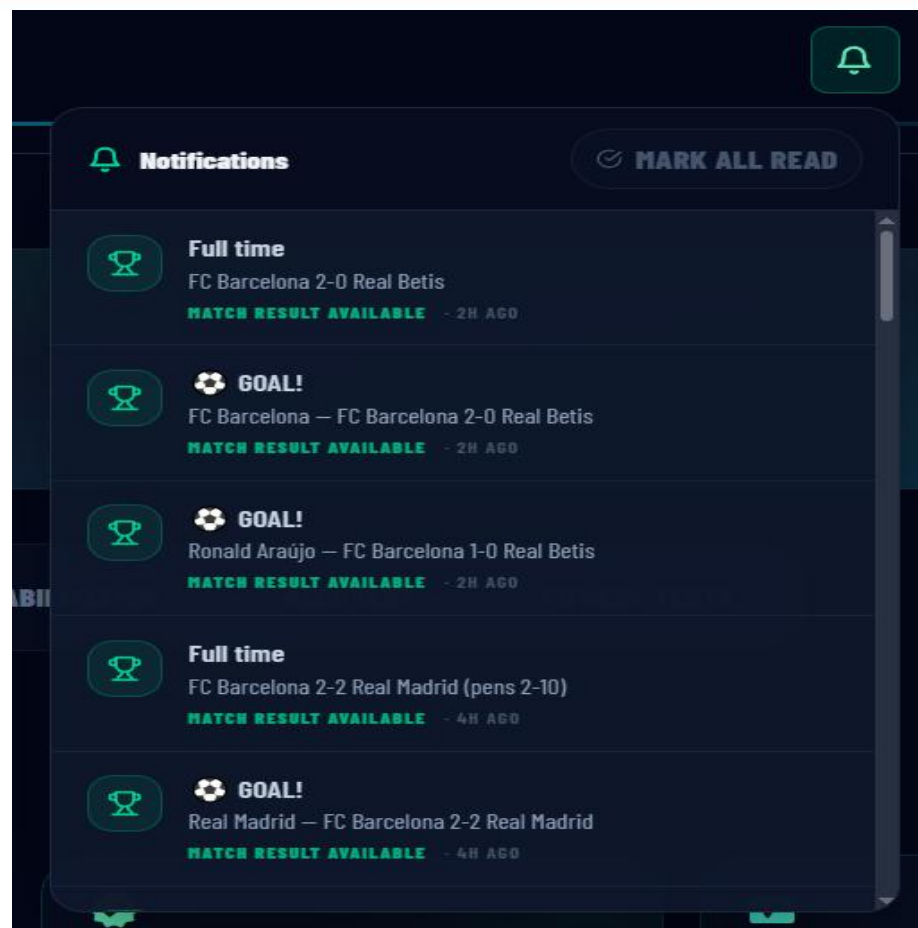


fig-notifications.png



Role Views

The same platform, a different surface for every login.

Administrator — what they see



Screenshot placeholder — drop `shot-view-admin.png` into `documentation/screenshots/`

Full access — every section, plus User Management.

shot-view-admin.png



Sport Manager — what they see



Screenshot placeholder — drop `shot-view-sportmanager.png` into `documentation/screenshots/`

Squad, staff, matches, scouting, contracts and sponsors.

shot-view-sportmanager.png

Head Coach — what they see



Screenshot placeholder — drop `shot-view-headcoach.png` into `documentation/screenshots/`

Training, line-ups, matches, medical and analytics.

shot-view-headcoach.png



Team Doctor — what they see



Screenshot placeholder — drop `shot-view-doctor.png` into `documentation/screenshots/`

The medical module, players and training availability.

shot-view-doctor.png



Scout — what they see



Screenshot placeholder — drop `shot-view-scout.png` into `documentation/screenshots/`

Scouting, prospects, matches and head-to-head.

shot-view-scout.png



Player — what they see



Screenshot placeholder — drop `shot-view-player.png` into `documentation/screenshots/`

A read-only view of club, matches, training and own medical.

shot-view-player.png



Fan — what they see



Screenshot placeholder — drop `shot-view-fan.png` into `documentation/screenshots/`

Spectator-only — club hub, trophies, competitions and matches.

shot-view-fan.png



LIVE DEMO

See Sportify in Action

A short walkthrough — login, the role dashboard, a live match, the medical room and team messaging.

Insert → Video here, or run the live app



Future Work

Where Sportify goes next: mobile, ML and the wider sporting world



Native Mobile Apps

Ships dedicated iOS and Android apps with push notifications so staff and players manage the club on the move.



Deeper ML Models

Extends the Python ML service with richer predictions fed by live in-match features, not just pre-game stats.



More Sports & Federations

Expands the multi-sport engine with new disciplines and official federation data feeds beyond football-data.org.



Wearables & Biometrics

Integrates wearables to stream training-load and biometric data straight into player medical and fitness profiles.



Live Video & Streaming

Attaches live video and clip streaming to the match timeline so every key event links to its footage.

✓ Conclusion

Sportify turns a club's scattered, manual operations into one secure, real-time, data-driven platform.

-  A production-grade microservices architecture, deployed with Docker.
-  Real competitions, a live-match engine and a complete medical workflow.
-  Role-based security end to end, with analytics and ML predictions on top.
-  An extensible foundation ready for mobile, more sports and richer models.



Team Members

Supervised by Dr. Shaimaa Saber



Mariam Mohamed Ahmed



Mohamed Abdellatief



Eman Ahmed Elboghdady



Khaled Mohamed



Shahd Abdelaziz



Mohamed Salem



Thanks for your attention!

Questions & discussion welcome

Sportify · MSCMS — a cloud-native multi-sport club management platform
